TABLE II
Values in Hex to be Loaded to Timer 1 (Oscillator Frequency 11.0592 MHz)

Notes	a	a#	b	c	c#	d	d#	e	f	f#	g	g#
Octave 4: A4=220 Hz	F7D1	F847	F8B6	F91F	F982	F9DF	FA37	FA8A	FAD9	FB23	FB68	FBAA
Octave 5: A5=440 Hz	FBE9	FC23	FC58	FC8F	FCC1	FCEF	FD1B	FD45	FD6C	FD91	FDB4	FDD5
Octave 6: A6=880 Hz	FDF4	FE12	FE2D	FE48	FE60	FE78	FE8E	FEA3	FEB6	FEC9	FEDA	FEEB
Octave 7: A7=1.76 kHz	FEFA	FF09	FF17	FF24	FF30	FF3C	FF47	FF51	FF5B	FF64	FF6D	FF75
Octave 8: A8=3.52 kHz	FF7D	FF44	FF8B	FF92	FF98	FF9E	FFA3	FFA9	FFAE	FFB2	FFB7	FFBB

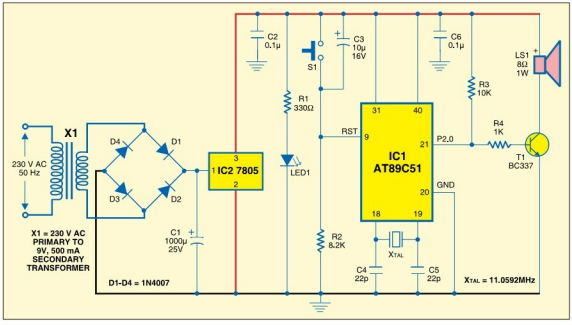


Fig. 3: Circuit of microcontroller-based ringtone generator

with octave 5 has a frequency of 440 Hz (refer Table I):

A5 = 440 Hz

So we get:

A6 = 880 Hz (A5x2)

A7 = 1760 Hz (A6x2)

The space between two consecutive octaves like A5 and A6 is divided into eleven equally spaced parts on the log-arithmic scale. Thus there are twelve equally spaced notes per octave: A5, A#5, B5, C5, C#5, D5, D#5, E5, F5, F#5, G5, G#5 and then A6 starts.

RTTTL is a simple text-based format that you can use to create ring tones. An RTTTL file is made up of a single string divided into three sections separated by colons (:).

In the example of a Happy Birthday song given below:

d=4, o=5, b=125:16c, 32p, 32c, 32p, 8d, 32p, 8c, 32p, 8f, 32p, e, 16p, 16c, 32p, 32c, 32p, 8d, 32p, 8c, 32p, 8g, 32p, f, 8p, 16c, 32p, 32c, 32p, 8c6, 32p, 8a, 32p, 8f, 32p, 8e, 32p, 8d, 32p, 16a#, 32p, 32a#, 32p, 8a, 32p, 8f, 32p, 8g, 32p

1. The first section is the name of the RTTTL melody, i.e., 'Happy Birthday Song.'

2. The second section defines the default values for the file. There are three categories of default values: duration (d=4), octave (o=5) and beat per minute (b=125).

3. The third section describes the melody.

It is a set of notes separated by a comma. The notes are given in the following format:

Duration (dn) Note Octave

8 c 6

where 'dn' is the duration of the present note.

If either the duration (dn) or octave is not specified for a particular note, the default values are assumed.

After getting the notes, you must calculate the duration for which a note is to be played:

Number of notes per second to be played (N) = 60/beats per minute (b)

=60/125

Net duration D in seconds for which a note is to be played = No. of notes per second × (Default note duration)/ (Duration specified for the current note)

= N×d/dn

= (60/125) ×(4/8)

= 30/125

So to play music, all you need to do is to get the RTTTL ring tone of the particular music, then read its note and generate the frequency for the calculated duration (D).

"If the note is dotted; for example, '2b' (here 'b' denotes note, not beats per second), the duration is made 1.5 times, so that D=1.5D."

Generation of PWM frequency. Since we are dealing with digital systems, we need to generate PWM (pulse-width-modulated) signals from port pin P2.0 of the microcontroller. A PWM signal consists of 'high'